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On-Board Diagnostics (OBD) for Euro Certified Engines

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Purpose

This service bulletin contains information regarding on-board diagnostics (OBD) for Cummins® engines certified to Euro 4, 4.5, and 5 emissions levels. The purpose of this bulletin is to define common OBD terms and explain the proper troubleshooting and fault code clearing strategies.

What is OBD?

OBD is a government-mandated standard that requires engines to actively monitor and test emissions-related components and systems to detect malfunctions that adversely affect emissions. An engine's OBD system monitors nearly every component that can affect the emission control system. If the OBD system detects a malfunction that could cause an increase in exhaust emission levels, the OBD system illuminates the Malfunction Indicator Lamp (MIL) on the vehicle instrument panel to alert the operator that the engine is in need of repair. The level of OBD monitoring required can vary depending on factors such as Gross Vehicle Weight Rating (GVWR), model year, certification level, and applicable government regulations.

OBD Terminology

To properly troubleshoot and diagnose Euro OBD-equipped engine systems, it is important to understand the following terms:

- **Fault Code (FC):** A code reported and stored by the engine ECM which indicates that a particular malfunction or abnormal condition has been detected. Different failure modes cause different fault codes to be stored, which provides direction for the appropriate troubleshooting and repair. Fault codes can be read by connecting to the ECM with a scan tool, such as the INSITE™ electronic service tool.
- **Malfunction Indicator Lamp (MIL):** A dash lamp that illuminates and alerts the operator when an OBD fault code becomes "Active", indicating an engine malfunction that could impact emissions.
- **OBD Monitor:** A diagnostic test or series of tests which are run by the engine ECM and are designed to determine the operational status of a specific emissions-related component or subsystem. OBD-equipped engines have multiple OBD monitors that run under certain operating conditions. These monitors test their respective systems and store or report the results accordingly.
- **Continuous Monitor:** A diagnostic that runs continuously during normal engine operation. It records a fault code and illuminates the MIL immediately after the diagnostic runs and does

not pass.

- Non-Continuous Monitor: A diagnostic that runs **only** under certain enabling conditions. A non-continuous diagnostic may run every time certain operating or environmental conditions are met, or once per drive cycle.
- OBD 1 Trip Fault: A fault code that is set to "Active" and illuminates the MIL after the corresponding diagnostic for the fault code runs and does **not** pass once during a drive cycle.
- OBD 2 Trip Fault: A fault code that is set to "Active" and illuminates the MIL after the corresponding diagnostic for the fault code runs and does **not** pass during two consecutive drive cycles.
- Drive Cycle: A specific series of steps or set of conditions that a vehicle **must** be operated under to enable a specific diagnostic to run. This can be part of the process required to clear certain OBD fault codes. Drive cycle conditions are stated in the troubleshooting tree for the applicable fault code.
- Ignition Cycle: The most common type of drive cycle, which begins with engine start and ends with engine shutdown.
- Warm-up Cycle: A drive cycle that includes an increase of at least 22.3°C [40°F] coolant temperature where the coolant temperature passes through 60°C [140°F]. The next warm-up cycle does **not** begin until the engine has been shut off, allowed to cool down to below 60°C [140°F], and is restarted.
- Cold Soak: A portion of certain drive cycles in which a vehicle **must** sit for a minimum of 8 hours with the engine off. This allows all temperature sensors to equalize at ambient temperature.
- Derate: An action caused by certain fault codes which decreases available engine power. This is done to protect the engine from damage and/or help initiate a service event. Some derates occur immediately, while others occur after a certain amount of time since a fault became "Active". Once the repair is made and the fault goes "Inactive", the engine will no longer be derated.
- Complete: The OBD Monitor has collected enough information to determine the health of its respective system.
- Not Complete: The OBD Monitor has **not** collected enough information to determine the health of its respective system. The fault completion status does **not** need to be considered during troubleshooting, and is **not** a fault code state. It is provided in the INSITE™ electronic service tool as information **only**.
- Pending: An OBD fault code becomes "Pending" once a diagnostic for an OBD 2 trip fault has run and **not** passed on one drive cycle. Fault code is "Active", but no dash lamps have been illuminated.
- Confirmed: An OBD fault code becomes "Confirmed" once the OBD system has gathered enough information to confirm that a malfunction exists and a fault code has become "Active". Note that an OBD fault can remain "Confirmed" even after the repair has been made and the MIL goes off. See the OBD Fault Code Functionality section of this bulletin for further details.
- Non-Erasable Fault Code: Fault code history can **not** be erased from the ECM memory with a scan tool such as the INSITE™ electronic service tool. Once the repair is made and the diagnostic has run, the MIL will go off, but the fault will remain "Inactive" and "Confirmed". Depending on how the fault code is configured, the "Inactive" fault is deleted from the fault

history after the successful completion of a set number of drive cycles, or after a set amount of time. These settings are defined by the regulatory agency and are programmed into the engine ECM.

Not all fault codes have the potential to impact emissions. Therefore, OBD-equipped engines can have both OBD and non-OBD fault codes. Typically, non-OBD fault codes illuminate either the Amber Warning Lamp (AWL) or Red Stop Lamp (RSL), which are the traditional Cummins® dash lamps. OBD faults always illuminate the MIL, and in some cases the AWL or RSL are illuminated as well.

Troubleshooting OBD Fault Codes

The preferred strategy for troubleshooting OBD fault codes is the same as for traditional Cummins Inc. fault codes: troubleshooting based on the Cummins Inc. fault status, as displayed in the "Fault Codes" INSITE™ electronic service tool screen. The "OBD Fault Codes" and "OBD Monitors" INSITE™ electronic service tool screens are for information **only** and should **only** be used for advanced troubleshooting.

During the troubleshooting process, the appropriate fault code troubleshooting tree for each fault code **must** be referred to in order to complete the repair. The troubleshooting trees can be found in the applicable Electronic Control System Troubleshooting and Repair Manual. Once a repair is made, the troubleshooting tree provides instructions on how to get the diagnostic to run on a drive cycle in order to validate the repair. If the repair was successful, the Cummins Inc. fault code status (which can be monitored in the "Fault Codes" INSITE™ electronic service tool screen) will become "Inactive" once the diagnostic runs and passes. This should be done for each fault code present in the ECM.

Extinguishing the MIL

Some fault codes require one drive cycle to extinguish the MIL, and some require three drive cycles. The differences are explained below:

One Drive Cycle MIL Off Fault Clearing

All non-erasable faults are "MIL off immediate faults", meaning the MIL will turn off immediately after the diagnostic runs and passes on one drive cycle. Since the "Inactive" fault can not be cleared with the INSITE™ electronic service tool, the repair is done at this point for non-erasable faults. Non-erasable faults are the only "MIL off immediate faults".

Three Drive Cycle MIL Off Fault Clearing

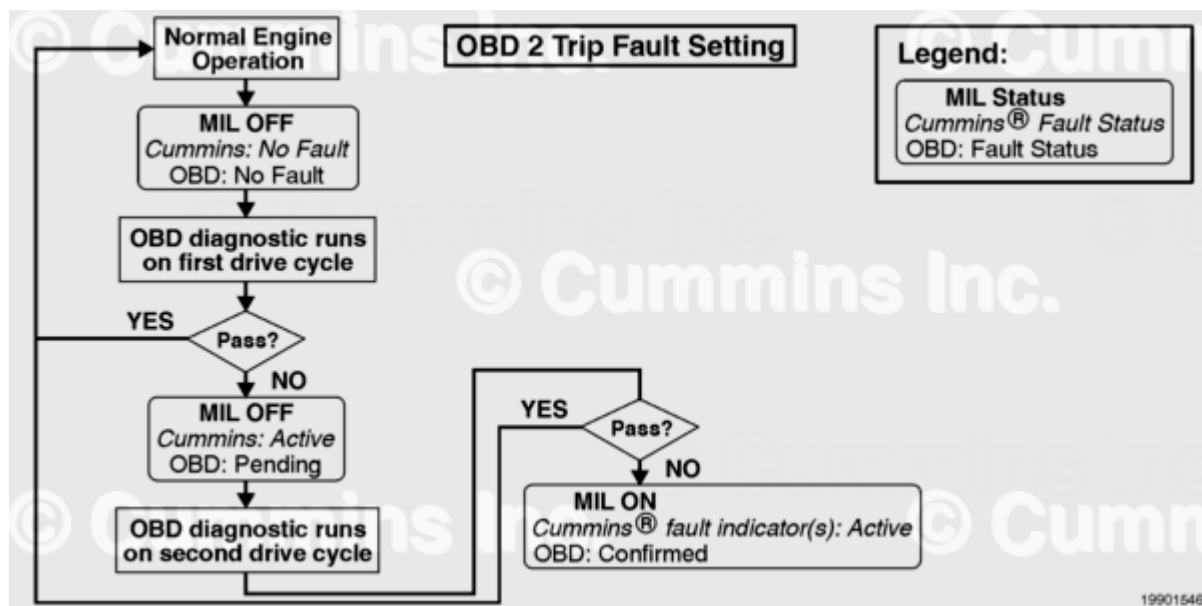
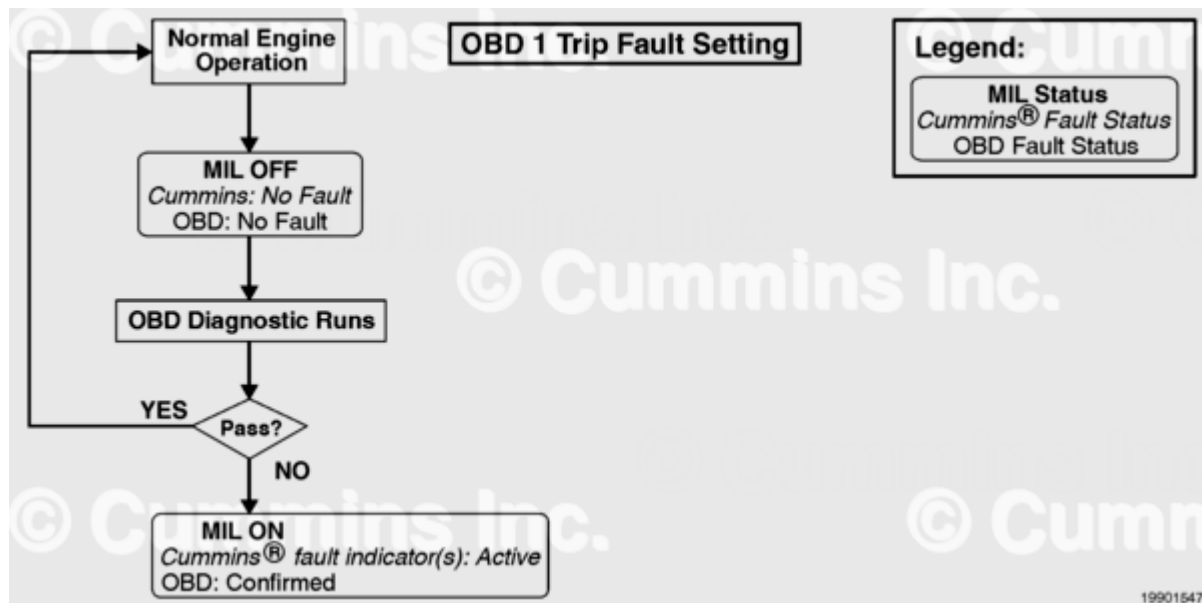
For the remainder of the OBD faults, the fault codes go "Inactive" after the diagnostic runs and passes once, but the MIL stays on until two additional drive cycles are completed in which the diagnostic runs and passes. When one drive cycle has been completed and the fault code is "Inactive", the repair has been validated, and the "Inactive" fault code can be cleared with the INSITE™ electronic service tool. This extinguishes all dash lamps. If the "Inactive" fault code is not cleared with the INSITE™ electronic service tool, the MIL will stay on until the diagnostic has run and passed on two additional drive cycles (three drive cycles total).

The troubleshooting trees provide important information, such as whether a fault code is non-erasable, how the fault code is set, what must be done to get the diagnostic to run, and how many drive cycles are required to turn the MIL off. For most fault codes, a drive cycle can be completed by starting the engine, letting it idle for 1 minute, and shutting it down. However, some fault

codes require that the vehicle be driven or operated on a chassis dynamometer in order to get the diagnostic to run and make the fault code go "Inactive".

OBD Fault Code Functionality

The following illustrations show the different ways that OBD fault codes are set to "Active" and the MIL is turned on:



The following illustrations show the different ways that OBD fault codes are cleared and how the MIL is turned off:

